

Applications of AI in Africa (*AI-Based System for Malaria Vector Surveillance in Africa Using Mosquito Wingbeat Frequency*)

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ABSTRACT

Malaria remains a major public health burden in Sub-Saharan Africa, where effective vector control is hindered by slow manual surveillance methods. Conventional entomological monitoring relies on trapping and expert identification, which is costly, restricted to fixed sites, and yields outdated data that impedes targeted resource allocation. This report presents an Artificial Intelligence (AI) system that identifies malaria-vector mosquitoes using the acoustic signatures of their wingbeats and maps each classification to its recording location. Using 6,008 labeled audio samples from the HumBugDB dataset, resampled to 8 kHz and converted into 40 Mel-frequency cepstral coefficients, we trained a hybrid architecture combining a 2D convolutional network, a bidirectional LSTM layer, and self-attention pooling. SpecAugment was applied to prevent the model from memorizing background noise. On a held-out test set of 395 recordings, the model achieved 84.6% overall accuracy, with 91.0% precision and 87.0% recall for the malaria-vector class. Zero-shot evaluation in 1,335 unseen field recordings from Kenya yielded 63.5% accuracy and 69.7% vector recall, demonstrating that the model learned generalized wingbeat features rather than recording artifacts. Finally, classifications are integrated into a web-based application that aggregates detections into a regional risk map to support operational vector control.

I. INTRODUCTION AND RELATED WORKS

A. Introduction

Malaria is a life-threatening infectious disease caused by Plasmodium parasites and transmitted to humans through the bite of infected female Anopheles mosquitoes. The disease is most prevalent in tropical and subtropical regions characterized by high humidity, warm temperatures, abundant rainfall, and agricultural activities that create stagnant water, providing suitable breeding habitats for mosquitoes [1]. Sub-Saharan Africa presents the most favorable climatic conditions and agricultural activities conducive to this, favoring the transmission of malaria.

According to the World Malaria Report 2025 of the World Health Organization (WHO), Africa represented approximately 95% of global malaria cases and deaths in 2024, with nearly a child under the age of five dying of malaria every minute [2] [3].

Despite significant progress in malaria control through interventions such as use of insecticide-treated mosquito nets, residual indoor spraying, and other vector-control strategies, malaria continues to pose a major public health challenge in many African countries.

B. Related Works

The severity and widespread impact of malaria have motivated researchers to develop approaches for its prevention, control, and early prediction. In Africa, malaria risk mapping commonly integrates spatial technologies such as remote sensing and geographic information systems (GIS) with environmental factors, routine epidemiological data, and reported malaria cases [4]. However, according to Gebreslasie [4] "there is growing recognition of the need to implement programs to predict and prevent malaria epidemics."

Machine learning (ML) offers a promising approach for analyzing mosquito-related data and supporting malaria surveillance. Mosquito wingbeat acoustics have been investigated for identifying mosquito species, determining sex, and analyzing behavioral characteristics [5], [6]. Because wingbeats produce characteristic acoustic patterns that can be used for mosquito classification.

Several studies have applied ML and deep learning techniques to mosquito wingbeat recordings. Convolutional Neural Networks (CNNs) have been widely used to learn discriminative patterns from acoustic representations such as Mel-spectrograms and Mel-Frequency Cepstral Coefficients (MFCCs) [7], [8]. For example, one study proposed a parameter-efficient cascaded framework that first separated mosquito sounds from background noise and subsequently classified six mosquito species using MFCC features, achieving an accuracy of 89% [7]. Another study investigated acoustic event detection for identifying mosquito wingbeat sounds in uncontrolled environments and achieved an ROC-AUC of 0.97 [8]. However, this approach focused on binary detection rather than identifying individual mosquito species. Collectively, these studies demonstrate the potential of acoustic features and ML techniques for automated mosquito classification.

Nevertheless, existing studies primarily focus on improving classification accuracy and model performance, with limited attention to integrating mosquito classification into practical malaria

surveillance systems. In particular, they generally do not incorporate the geographic location of recorded mosquitoes to support region-level malaria risk assessment. This creates an opportunity to extend mosquito acoustic classification beyond species identification toward a surveillance application that can provide geographically relevant information to public health institutions.

This project builds on previous work by developing an AI-based malaria surveillance system that classifies mosquito wingbeat recordings as malaria-vector or non-malaria-vector and associates the classification results with the geographic location of the recordings. The system is intended to provide timely information that can support public health institutions in identifying areas requiring further malaria-risk assessment and intervention. The proposed use of a web app as an acoustic sensor also offers a potentially affordable and accessible approach for collecting mosquito recordings in field environments [9].

II. PROBLEM STATEMENT

Although global malaria case incidence declined by 25.6% between 2000 and 2015 (from 79.4 to 59.0 cases per 1,000 population at risk), progress has stalled and partially reversed. Between 2015 and 2024, malaria case incidence increased by 8.5%, rising from 59.0 to 64.0 cases per 1,000 population at risk [2].

This resurgence highlights the need for effective surveillance methods capable of monitoring mosquito vector abundance to support the early identification of areas at elevated risk of malaria transmission.

III. METHODOLOGY

A. Data Preprocessing and Feature Extraction

An open-source multi-species dataset of 6008 acoustic recordings of mosquitoes from indoor and outdoor background environments from Tanzania, Thailand, Kenya, the USA and the UK was used. We removed unknown species, filtered audio recordings, and organized data for efficient training.

Binary classes were defined; 0: 'non-vector' class with *Culex pipiens* complex, and *Aegypti*, and 1: 'malaria-vector' class combining 4 primary malaria-vector i.e. *arabiensis*, *funestus* ss, *squamosus*, and *coustani*. Acoustic features, such as harmonics and spatial patterns that differentiate mosquito species, were extracted to be used in ML model training.

Audios were optimized for a sample rate of 8000Hz, then padded or truncated and converted to tensors to ensure uniform tensor dimensions.

B. Model Development

A 2D-CNN-BiLSTM architecture with Self-Attention Pooling (MosquitoAttnNet) was developed. The model extracts 40 Mel-Frequency Cepstral Coefficients (MFCCs) directly from raw audio waveforms. During training, a SpecAugment pipeline applies frequency and time masking to prevent the network from memorizing static background noise. The acoustic representations are processed through a 2D Convolutional backbone, mapped into a 2-layer Bidirectional LSTM, and aggregated using Self-Attention Pooling to dynamically emphasize mosquito flight tones over ambient silence. The network was optimized using AdamW (learning rate = 0.0005, weight decay = 1×10^{-3}), dynamic class penalty weights, and a ReduceLROnPlateau scheduler over 40 epochs.

C. Web Application

A web app-based user interface was developed for proof of concept, where users can upload mosquito audios, view prediction results for the classification of mosquito species, and visualizing mosquito surveillance information (risk rate of each region, and more).

D. Risk Mapping

Use model results to map data points to the region where the recordings belong. Determine the risk rate of malaria spread for each region using hotspots on the map.

E. List of Assumptions

- 1) The presence of a high rate of female *Anopheles* mosquitoes in a region is an indicator of the potential risk of malaria transmission.
- 2) The considered malaria-vector species are all 'female' *anopheles* with potential to transmit malaria.
- 3) The audio data in the HumBugDB acoustic mosquito dataset is sufficient and reliable to train an ML model accurately.
- 4) Accurate performance of the model on Kenyan data implies that model will work well anywhere else in Africa.

F. List of Resources

Dataset: HumBugDB acoustic mosquito dataset from Zenodo [10].

Programming Language: Python

Machine Learning Architecture: CNN

Other Tools: VSCode, Google Colab, GitHub

IV. RESULTS AND DISCUSSION

A. Result

The system developed is divided into two parts: An ML model that performs binary classification and a web application with which the user will interact with the system. The process starts with the web app.

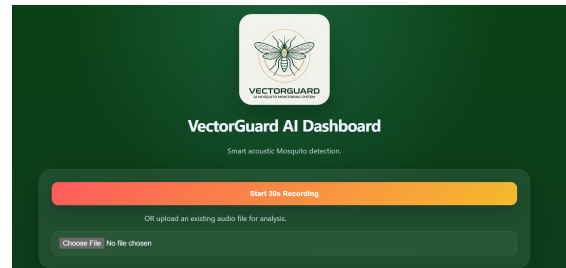


Fig. 1: web app front page.

The user will either record live audio from the area under surveillance by clicking "Start 30s Recording" or upload a pre-recorded audio.



Fig. 2: Analyze button.

After recording or uploading an audio, the user will click "Analyze Acoustic Data" to run the ML model on the uploaded audio.

The app then returns an inference statistic of the audio in the form of a pie-chart showing how confident the model is when classifying the audio as either a malaria-vector or non-malarial vector.

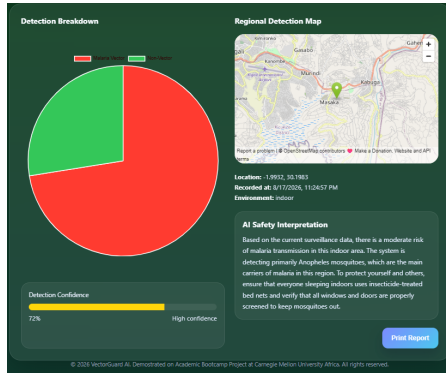


Fig. 3: Model Prediction Confidence and Localization

For every classification, the system populates the audio datapoint onto a map in order to create a heatmap of the malaria-vector concentration within Kenya, as a proxy country. The heatmap will then inform the user of regions that need the most attention due to the high presence of malaria-vectors to make the most rational and efficient resource distribution.

B. Discussion

Training Behavior

The model was trained for 40 epochs using AdamW with explicit weight decay (1×10^{-3}), dynamic class penalty weights, and SpecAugment regularization (Figure 4). The V6 model maintained a stable validation loss bounded between 0.34 and 0.36. The optimal model checkpoint was saved at Epoch 23, with validation accuracy peaking at 85.03% at Epochs 21 and 37.

Classification Performance

The confusion matrix on the held-out test set (395 samples: 118 non-vector, 277 malaria vector) is shown in Figure 5. The per-class classification metrics are summarized in Table I.

TABLE I: Per-class classification metrics on the test set.

Class	Precision	Recall	F1-Score
Non-Vector	72.0%	80.0%	76.0%
Malaria Vector	91.0%	87.0%	89.0%

The model achieved an overall accuracy of **84.6%**, correctly classifying 334 of 395 test samples. Vector detection precision reached **91.0%**, while vector recall reached **87.0%** (detecting 240 of 277 true vectors), reducing false negatives to 37 samples.

Zero-Shot Cross-Geographic Validation (Kenya Demo Dataset) To evaluate robustness against acoustic domain shift, the model was tested on 1,335 unseen field recordings from Kenya (598 Non-Vector, 737 Malaria Vector). The V6 architecture achieved **63.5%** zero-shot accuracy and a malaria vector recall of **69.7%** (detecting 514 of 737 unseen vectors), confirming that the 2D-CNN and attention mechanism extract generalized biological wingbeat features rather than recording-specific noise artifacts.

V. LIMITATIONS

- 1) Our solution depends on network connectivity, limiting deployment in remote areas.

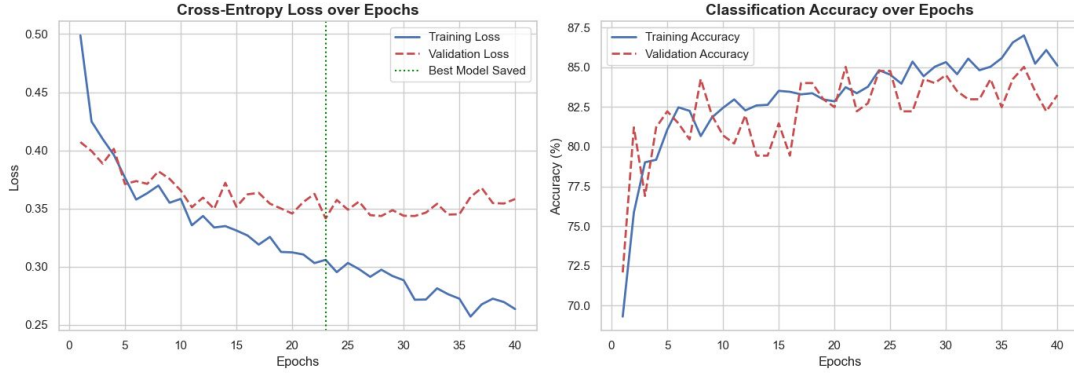


Fig. 4: Training and validation cross-entropy loss (left) and classification accuracy (right) over 40 epochs. The green dotted line denotes the best checkpoint saved at Epoch 23 (Validation Loss: 0.3415).

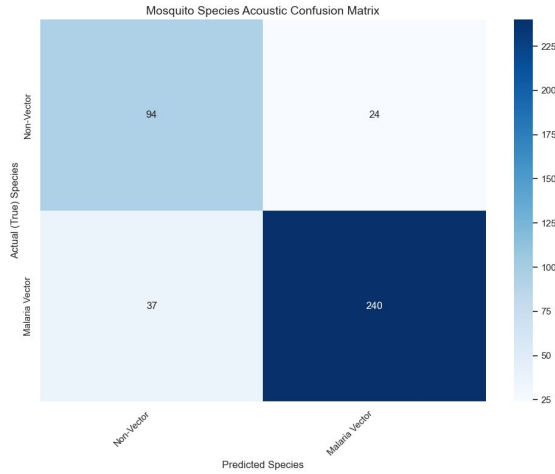


Fig. 5: Confusion matrix for acoustic classification of Malaria Vector vs. Non-Vector mosquito species on the test set (395 samples).

VI. CONCLUSION AND RECOMMENDATIONS

A. Conclusion

This project developed an AI-based acoustic system to classify malaria-vector mosquitoes from wingbeat recordings. An app was built for field surveillance, supporting audio recording, inference, and visualization of results, including a heatmap of malaria-vector concentration [11]. Further training data and tuning would improve reliability for more decision-making use.

B. Recommendations

- 1) Quantize and export the model to edge runtime formats to enable fully offline, on-device web inference.
- 2) Acoustic domain shift across diverse geographical sites can introduce variance in zero-shot transfer performance.
- 3) High-frequency temporal resolution in noisy background environments still requires continuous acoustic exposure.
- 2) Use dynamic classification threshold tuning during live deployment to maximize detection sensitivity in high-risk transmission zones.
- 3) Deploy dedicated standalone acoustic IoT hardware nodes with automated low-power wake-on-sound triggers.

VII. AI USE DISCLOSURE

A. *AI was used to polish the language used in some parts of the report for sentence structure and word choice.*

B. *AI was utilized in debugging code.*

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